SUPPORTING ADULT GUIDANCE

Beyond the point: tenths, hundredths, thousandths

Place Value · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Decimal Zoom, learners tap a gap on a number line from 0 to 1. The line zooms into that tenth, showing ten hundredths, and then into a hundredth, showing ten thousandths. A place-value chart (Ones . Tenths Hundredths Thousandths) builds the number as they go. Find buttons set targets such as 0.374, 0.908 and 0.013; a wrong gap is refused with a message naming the digit to look at. Six questions follow, and a stretch explains why the column after the ones must be tenths. Learners then write amounts as decimals, judge statements as true or not, and make decimals that fit descriptions.

Driving Question: *What lives to the right of the decimal point?*

The mathematics, in plain terms

Place value is the same on both sides of the point: each column is worth one tenth of the column to its left, so the columns after the ones are 1/10, 1/100 and 1/1000. A decimal such as 0.374 can be read as 3/10 + 7/100 + 4/1000, or as 374/1000. Trailing zeros after the point do not change the value (0.5 = 0.50). Ordering decimals is taught in Lesson 6; one statement here (0.3 and 0.25) previews it.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Zooming in to 0.374

Where does 0.374 live on a number line from 0 to 1?

Zoom 1: 0.374 is between 0.3 and 0.4

Ones		Tenths	Hundredths	Thousandths
0	•	3		

Zoom 2: between 0.37 and 0.38

Ones		Tenths	Hundredths	Thousandths
0	•	3	7	

Zoom 3: found 0.374

Ones		Tenths	Hundredths	Thousandths
0	•	3	7	4

Each zoom splits one gap into ten, and adds one digit after the point.

1. The tenths digit is 3.	between 0.3 and 0.4
2. Zoom in. The hundredths digit is 7.	between 0.37 and 0.38
3. Zoom in again. The thousandths digit is 4.	0.374

0.374 = 3 tenths + 7 hundredths + 4 thousandths, which is 374 thousandths.

Example 2 — An empty column

Write 4 tenths and 9 thousandths as a decimal.

4 tenths and 9 thousandths

Ones		Tenths	Hundredths	Thousandths
0	•	4	0	9

The highlighted 0 holds the hundredths column, so the 9 stays in the thousandths.

1. Tenths digit	4
2. Hundredths: none, so hold the column with	0
3. Thousandths digit	9

0.409. Without the 0 it would be 0.49, which is 4 tenths and 9 hundredths.

Zoom first, then name the columns, then use them

Learners see a tenth split into hundredths, and a hundredth into thousandths, before naming any columns, so the names describe something they have watched. The stretch connects this to the × 10 and ÷ 10 movement from Lesson 2, so the decimal columns are a continuation rather than a new rule.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Decimal Zoom and six questions. Stretch: why must the column after the ones be tenths?	~16 min
Movement	Say it, write it: amounts as decimals.	~7 min
Reflection	True or not? Six statements about decimals.	~7 min
Creativity	Make it: decimals that fit a description.	~10 min
Rest	Zoom in, and there is always more room.	~2 min
Nutrition	Intellectual: precision.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading 0.07 as seven tenths.
Ask: “Which column is the 7 in? What is the column straight after the point?”

Thinking a longer decimal is bigger (0.25 > 0.3).
Ask: “How many hundredths is 0.3? And 0.25?”

Thinking a hundredth is bigger than a tenth, because a hundred is bigger than ten.
Ask: “Into how many parts is the whole split for tenths? For hundredths? Which parts are smaller?”

Writing 4 tenths and 9 thousandths as 0.49.
Ask: “Which column is the 9 in? What holds the hundredths?”

Saying “point twenty-five” for 0.25 and so losing the column values.
Ask: “Can you say it as tenths and hundredths?” (Reading the digits one at a time, “point two five”, keeps the columns visible.)

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: dividing by 10 (Lesson 2) and zero as a placeholder (Lesson 3). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 0.3 is smaller than 0.25 because 3 is smaller than 25. The aim is a question that brings the columns back into view.

- “Can you find both numbers on the zoom?”

- “Which tenth is each one in?”
- “How many hundredths is 0.3?”
- “What does the 2 in 0.25 tell you?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners say “point twenty-five”?
Encourage “point two five”. Reading digit by digit keeps each digit's column in view.

Why stop at thousandths?
Thousandths are enough to see that the pattern continues; the zoom shows learners it could go on for ever.

Is the stretch required?
No. The secure core is reading and writing tenths, hundredths and thousandths and saying what each digit is worth.

What comes next?
Lesson 5: many ways to split a number (partitioning).

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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